

Acupuncture visit frequency and its associations with quality of life, and cardiovascular parameters among hypertension outpatients: A cross-sectional study

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Introduction

Hypertension is a major global public health concern and an important risk factor for cardiovascular disease. Acupuncture has increasingly been used as a complementary therapy for hypertension, yet its associations with quality of life (QoL), nutritional anthropometry, handgrip strength (HGS), and blood pressure remain insufficiently understood. This study aimed to determine the associations between acupuncture visits and QoL, nutritional anthropometry, HGS and blood pressure among hypertension outpatients, with additional age-stratified analyses to examine differences between younger and older participants.

Materials and Method

Study Area: Traditional and Complementary Centre, UTAR

Sample Size: 124 participants

Inclusion Criteria: Outpatients who visited acupuncture at T&CM and clinically diagnosed with hypertension

Data Collection Tools: 1) Short Form-36 (SF-36) questionnaire

2) Automated Oscillometric Blood Pressure Monitoring

3) Nutritional Anthropometry Measurement (Stadiometer & Karada Smart Body Composition Analyzer)

4) Handgrip Strength (Dynamometer)

Data Analysis: SPSS version 31

Results

1 Table 1: Participant Characteristics

Variable	Findings
Participants	124 hypertension outpatients (response rate= 96.9%)
Acupuncture visit status	Firsttime (n=,30.6%) Non-firsttime (n=,69.4%)
Acupuncture frequency	Once a week (n=,21.0%) Once every two weeks (n=,25.0%) Once a month (n=,11.3%) Less than once a month (n=,12.1%) Not applicable/First-time (n=,30.6%)
Reasons for acupuncture	Pain relief (n=,49.2%); General wellness & energy improvement (n=,45.2%) Hypertension and diabetes management (n=,5.6%)

Key findings: 69.4% were non-first-time acupuncture users, while 30.6% were first-time users. Among all participants, the most common acupuncture frequency was once every two weeks (25.0%), followed by once weekly (21.0%). The main reasons for seeking acupuncture were pain relief (49.2%) and improving general wellness and energy levels (45.2%).

2 Table 2: Association between acupuncture visits and quality of life (QoL)

QoL Domain	Firsttime	Non-firsttime	<i>p</i> -value
Physical Functioning	56.7 ± 26.6	66.5 ± 24.3	0.049*
Mental Health	51.2 ± 13.6	57.6 ± 13.7	0.013*

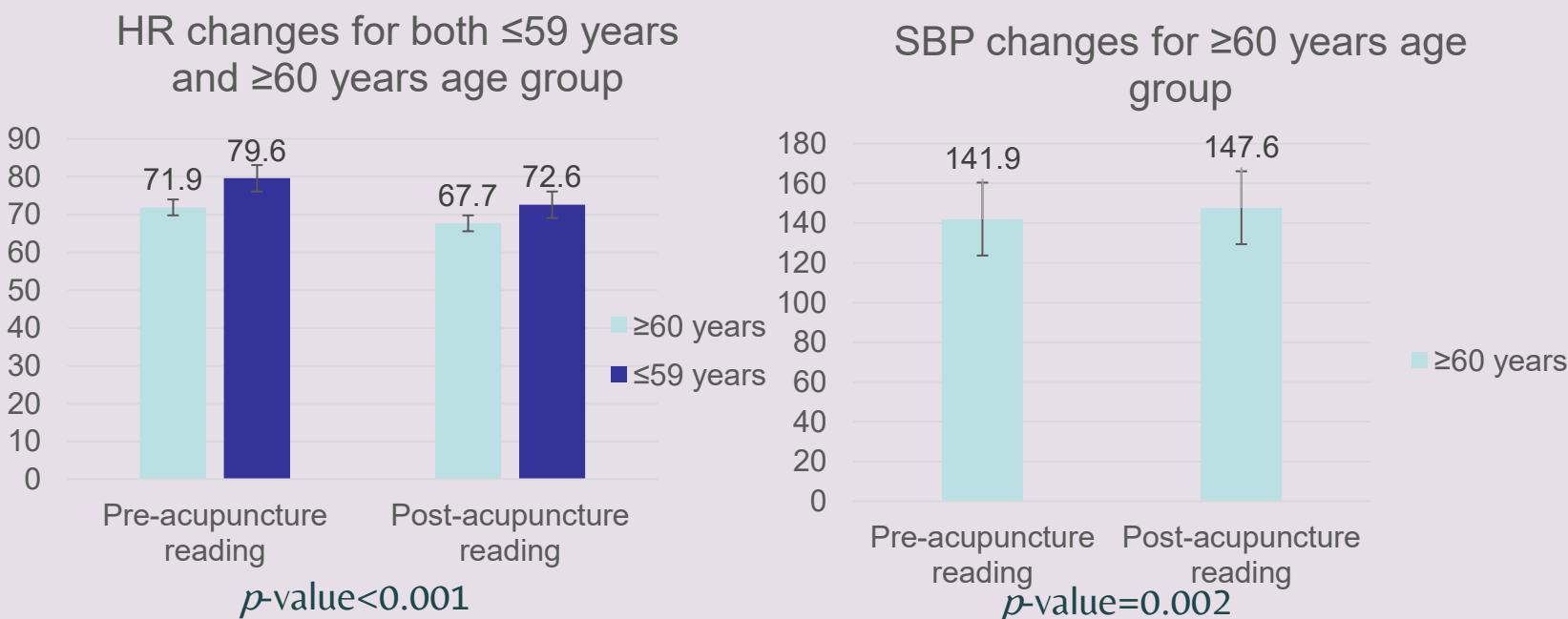
Key findings: repeat acupuncture users had significantly better physical functioning and mental health than first-time users

3 Table 3: Association between acupuncture visit and nutritional anthropometry and handgrip strength

Variable	Firsttime	Non-firsttime	<i>p</i> -value
Body Mass Index (kg/m ²)	27.72 ± 4.74	26.95 ± 4.76	0.491
Body Fat (%)	29.60 ± 7.72	28.99 ± 6.53	0.535
Visceral Fat	11.16 ± 5.64	10.41 ± 6.22	0.488
Subcutaneous Fat (%)	22.80 ± 8.04	23.00 ± 7.46	0.749
Skeletal Muscle (%)	28.82 ± 3.91	28.72 ± 3.74	0.837
Handgrip Strength (kg)	22.42 ± 6.19	22.73 ± 8.30	0.987

Key findings :No significant associations were observed between acupuncture visits and nutritional anthropometry or handgrip strength (*p* > 0.05).

4 Table 4: Age-stratified cardiovascular response following acupuncture



Key findings: SBP increased significantly after acupuncture among participants aged ≥60 years (*p* = 0.002). Heart rate decreased significantly after acupuncture in both age groups (*p* < 0.001).

Discussion/Conclusion

- Repeat acupuncture users reported significantly better Physical Functioning and Mental Health than first-time users, consistent with previous evidence suggesting potential benefits of acupuncture for physical symptoms and mental well-being (Fan *et al.*, 2024; Rodrigues *et al.*, 2023).
- Acupuncture visit status was not significantly associated with nutritional anthropometry, handgrip strength, SBP, DBP, or HR. Previous studies have reported improvements in BMI and body composition mainly following structured, longer-term acupuncture interventions, which may explain the lack of association by visit status in the present study (Zhao *et al.*, 2024).
- SBP increased in participants ≥60 years, contrasting with previous reports of reduced SBP after acupuncture (Sudhakaran, 2020), possibly due to differences in assessment timing. HR decreased in both age groups, suggesting a short-term autonomic response (Shen *et al.*, 2024).
- Further longitudinal studies are required to determine whether these associations represent causal effects of acupuncture.

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Declaration of conflict for all authors

The authors declare no conflict of interest.

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